

Internship Report
Of
Data Science Master Virtual Internship

Submitted

To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 25SCS1003003067

Name of Student: PALLAVI BORA

Batch: 2025-29

2026-27

CANDIDATE'S DECLARATION

I, Pallavi Bora, do hereby declare that the internship report titled **“Data Science Master Virtual Internship”** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Pallavi Bora

Roll No.: 25SCS1003003067

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Student Name: Pallavi Bora

Roll No.: 25SCS1003003067

INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Pallavi bora

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS

Gaurav Pai
Head - Strategy and Operations
Siemens Industry Software (India) Pvt. Ltd

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
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Certificate ID : 40d38f954444b77d6aeb

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GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a summary of the Data Science Master Virtual Internship undertaken as part of the AICTE – EduSkills National Internship Portal initiative, supported by Siemens Industry Software (India) Pvt. Ltd. The internship was completed over an 8-week duration from June 2026 to August 2026 while pursuing the Bachelor of Technology programme in Computer Science and Engineering at IILM University, Greater Noida. The programme was designed to provide industry-relevant, hands-on exposure to the core concepts and tools used in modern data science and machine learning workflows, combining structured weekly learning modules with practical assessments and a project-based approach. The internship aimed to bridge the gap between academic learning and industry practice by introducing learners to real-world data engineering, machine learning, and platform administration skills using the RapidMiner AI Hub and associated Siemens data science tools.

4.2 Organization Profile

The internship was supported by Siemens Industry Software (India) Pvt. Ltd., a global leader in industrial software and digital transformation solutions, and was delivered through EduSkills, an AICTE-recognized skilling partner working under the “Nation Building Through Skills” initiative. The programme was hosted on the AICTE National Internship Portal, a Ministry of Education initiative connecting students across India with structured virtual internship opportunities from leading industry partners. EduSkills is responsible for curriculum design, learner assessment, and issuance of the final internship credential, while Siemens provides the underlying technology platform (RapidMiner) and domain expertise in applied data science.

4.3 Problem Statement

Organizations today generate large volumes of data but often lack skilled professionals capable of transforming this data into actionable business insight using structured, repeatable machine learning workflows. The problem addressed by this internship was to develop the practical competencies required to identify suitable business problems for AI/ML, access and prepare data, build and evaluate machine learning models, and

deploy and administer these models on an enterprise-grade platform — skills that are directly transferable to real industry data science roles.

4.4 Project Objectives

- To understand how business problems are identified and framed as data science and machine learning use cases.
- To learn techniques for accessing, loading, transforming, and calculating data using industry-standard tools.
- To build foundational and advanced machine learning models and evaluate their performance.
- To master advanced data preparation and automation techniques for handling real-world datasets.
- To learn how machine learning models are deployed, monitored, and maintained in a production-like environment.
- To gain hands-on experience configuring and administering the RapidMiner AI Hub platform.

4.5 Scope of the Project

The scope of this internship was limited to the structured, week-wise virtual curriculum defined by Siemens and EduSkills under the “Data Science Master” track. It covered the complete lifecycle of a data science project — from problem identification and data engineering, through model building and evaluation, to deployment and enterprise platform administration — using simulated business scenarios and datasets provided within the RapidMiner learning environment. The internship did not involve deployment on a live external client project but focused on building strong conceptual and applied skills through guided modules, weekly assessments, and a final assessment test.

4.6 Technologies and Tools Used

- RapidMiner Studio and RapidMiner AI Hub – for data preparation, model building, and platform administration.
- Data Engineering tools – for accessing, loading, and transforming structured and unstructured data.

- Machine Learning algorithms – including classification, regression, and clustering techniques covered across the professional and master modules.
- AICTE EduSkills Virtual Internship Portal – for weekly assessments, project documentation submission, and the final credential evaluation.

4.7 System Architecture

The learning environment followed a typical enterprise data science pipeline architecture consisting of four broad layers:

- (1) a Data Access and Ingestion layer for connecting to and loading raw data sources;
- (2) a Data Preparation layer for cleaning, transforming, and enriching data;
- (3) a Modeling layer where machine learning algorithms are trained, validated, and tuned; and
- (4) a Deployment and Administration layer, implemented through the RapidMiner AI Hub, where trained models are published, monitored, and managed at an enterprise level. Each week of the internship corresponded to hands-on practice with one or more of these architectural layers.

4.8 Methodology

The internship followed a structured, week-wise learning methodology as prescribed by Siemens and EduSkills, combining self-paced video modules, hands-on exercises, weekly assessments, and a final assessment test. The week-wise breakdown of the methodology followed was as below:

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learned to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learned to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Built foundational machine learning skills.
Week 4	Data Preparation Master	Mastered advanced data handling and automation.
Week 5	Machine Learning Master	Learned advanced machine learning techniques.

Week 6	Applications & Use Cases Master	Deployed and monitored ML models.
Week 7	Platform Administration Master	Configured and managed RapidMiner AI Hub.
Week 8	Platform Administration Master	Performed enterprise-level platform operations.

Table 4.1: Week-wise structure of the Data Science Master Virtual Internship

In the final week, a Final Assessment Test was undertaken, and the overall grade was based on cumulative performance across weekly assessments, project submissions, and the final test, in accordance with the internship evaluation policy of AICTE-EduSkills.

4.9 Expected Outcomes

- Developed a strong conceptual understanding of the end-to-end data science and machine learning workflow.
- Gained practical, hands-on skills in data engineering, model building, and model deployment using RapidMiner.
- Acquired platform administration skills necessary for managing enterprise AI/ML solutions.
- Successfully completed all weekly assessments and the Final Assessment Test, earning the Virtual Internship Certificate.
- Strengthened employability and confidence in applying data science concepts to real-world business problems.

4.10 Certificates of Completion and Other Communication Proof

The Certificate of Virtual Internship issued jointly by AICTE, EduSkills, and Siemens, along with the official Internship Offer Letter (Ref. No. C17Y2026M081503620), are attached in this report as proof of successful completion of the Data Science Master Virtual Internship. The certificate confirms an overall grade of “E (Excellent)” for the 8 week programme conducted between June and August 2026.

BIBLIOGRAPHY / REFERENCES

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